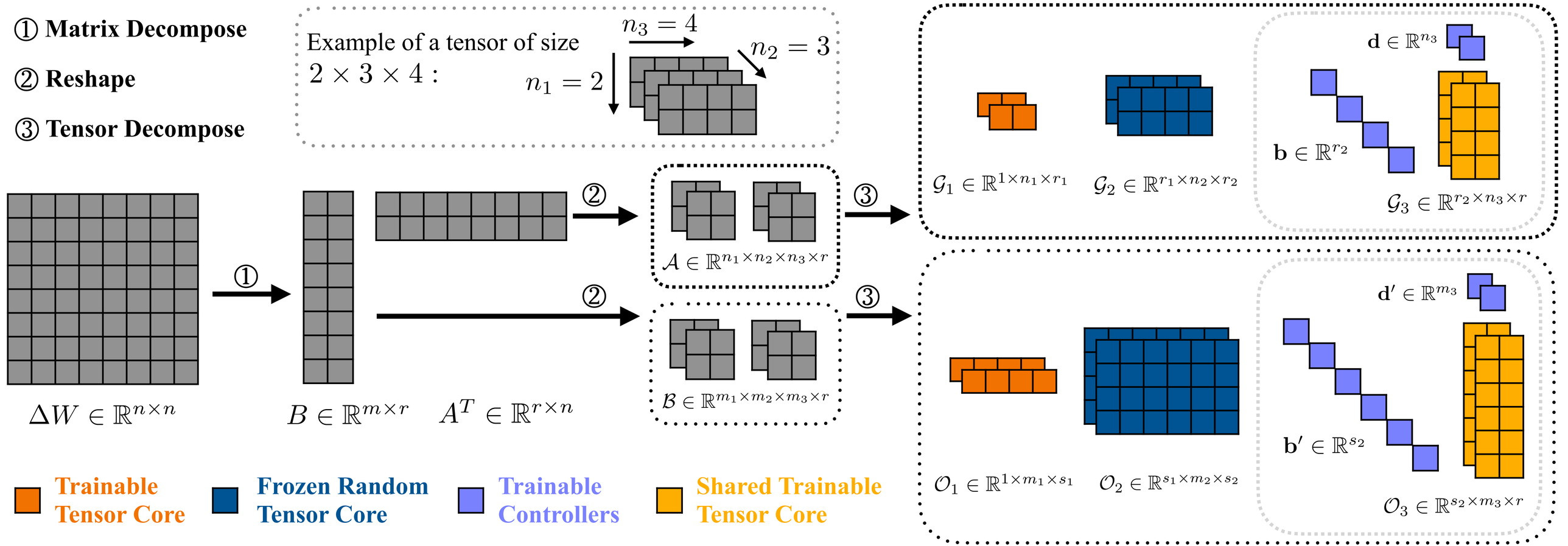
① Matrix Decompose
② Reshape
③ Tensor Decompose
Example of a tensor of size $2 \times 3 \times 4$:
$n_3 = 4$
$n_2 = 3$
$n_1 = 2$
$\Delta W \in \mathbb{R}^{n \times n}$
$B \in \mathbb{R}^{m \times r}$
$A^T \in \mathbb{R}^{r \times n}$
$\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3 \times r}$
$\mathcal{B} \in \mathbb{R}^{m_1 \times m_2 \times m_3 \times r}$
$\mathcal{G}_1 \in \mathbb{R}^{1 \times n_1 \times r_1}$
$\mathcal{G}_2 \in \mathbb{R}^{r_1 \times n_2 \times r_2}$
$\mathbf{d} \in \mathbb{R}^{n_3}$
$\mathbf{b} \in \mathbb{R}^{r_2}$
$\mathcal{G}_3 \in \mathbb{R}^{r_2 \times n_3 \times r}$
$\mathcal{O}_1 \in \mathbb{R}^{1 \times m_1 \times s_1}$
$\mathcal{O}_2 \in \mathbb{R}^{s_1 \times m_2 \times s_2}$
$\mathbf{d}' \in \mathbb{R}^{m_3}$
$\mathbf{b}' \in \mathbb{R}^{s_2}$
$\mathcal{O}_3 \in \mathbb{R}^{s_2 \times m_3 \times r}$
Trainable Tensor Core
Frozen Random Tensor Core
Trainable Controllers
Shared Trainable Tensor Core